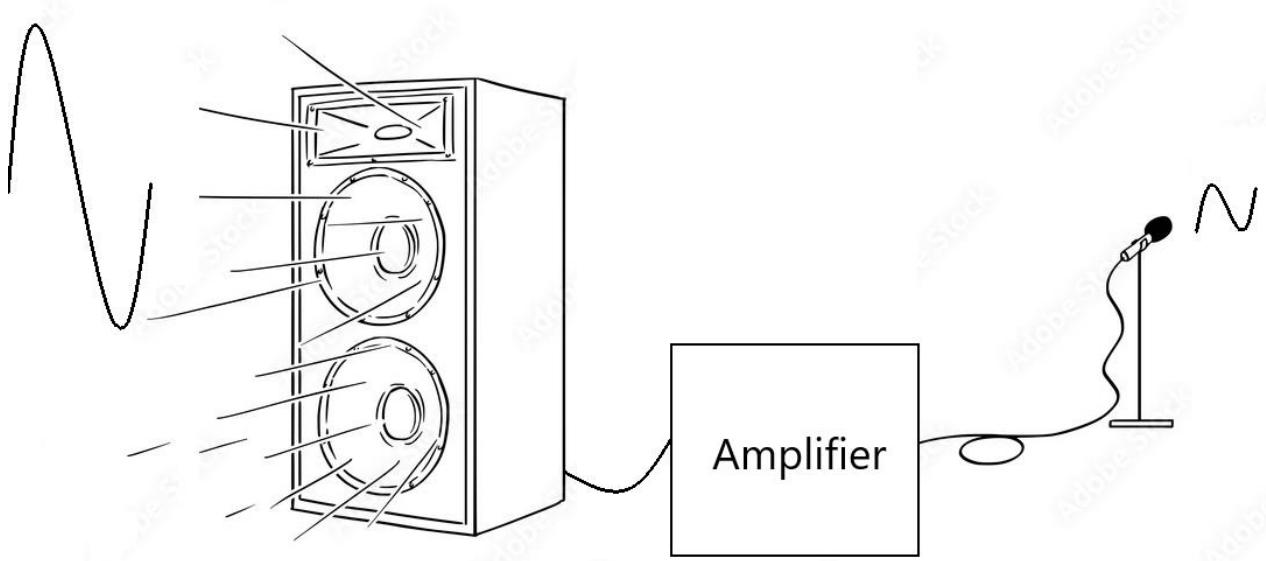
EE 312

Experiment 5

Operational Amplifiers

OP AMPS

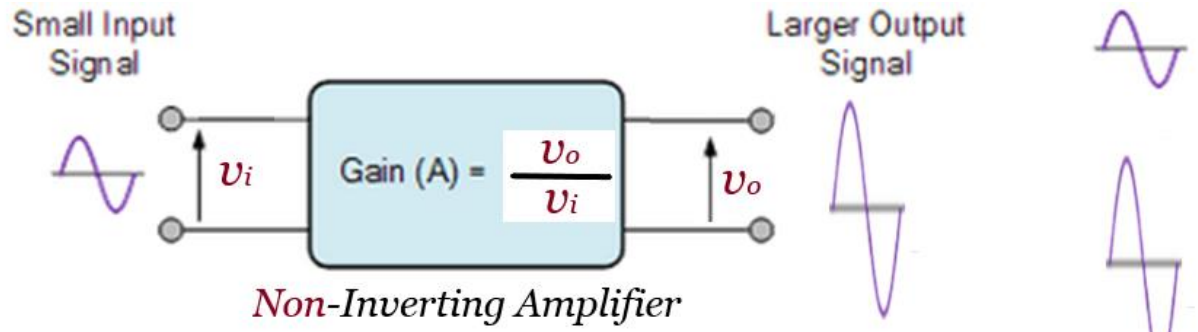
Turki Almadhi, Lecturer, Dept. of Elect. Engineering, KSU

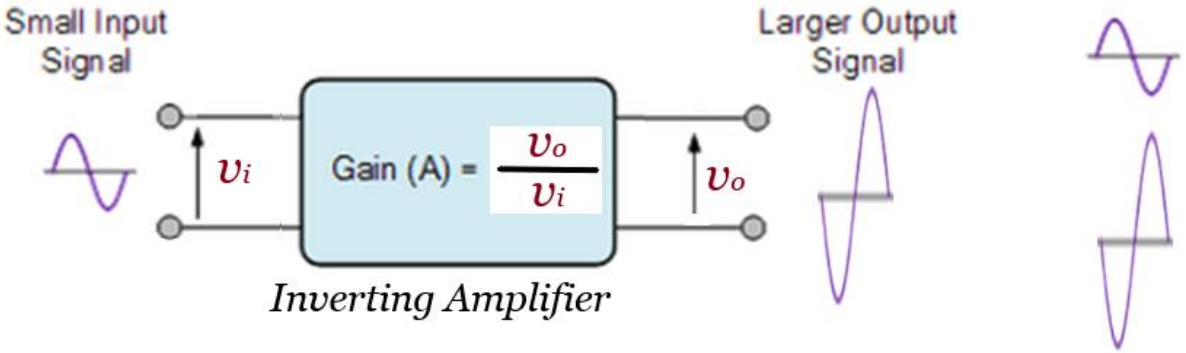


What is a Voltage Amplifier?

It is an electronic circuit which amplifies its output voltage relative to its input voltage.

Non-inverting Amp. *vs.* Inverting Amp.

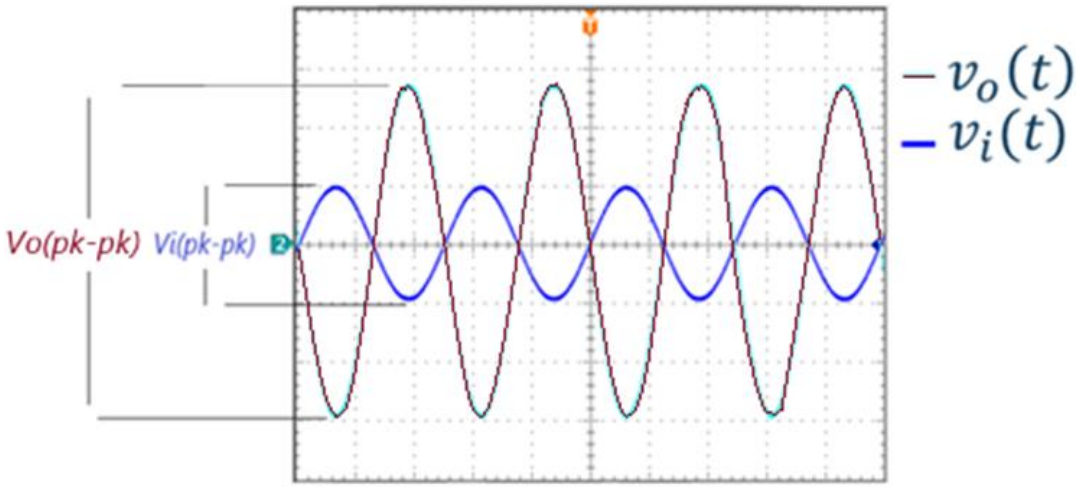




How to Measure Voltage Gain *A*

VOLTAGE GAIN A

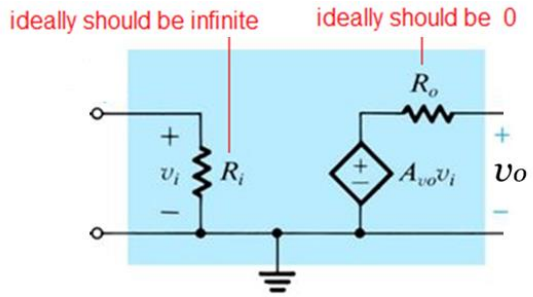
$$A = \frac{v_o(t)}{v_i(t)} = \frac{Vo(pk\text{-}pk)}{Vi(pk\text{-}pk)}$$



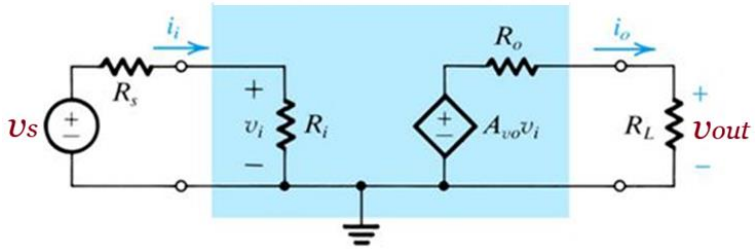
Intrinsic Voltage Gain *Avo* vs. Overall Voltage Gain *G*

- Due to Ri of the Amplifier not Being Infinite, Some Voltage is Lost Upon Connecting a Voltage Signal Source to the Amplifier.

- Due to Ro of the Amplifier not Being Zero, Some Voltage is Lost Upon Connecting a Load to the Amplifier.



Circuit model for the voltage amplifier.



The voltage amplifier with input signal source and load.

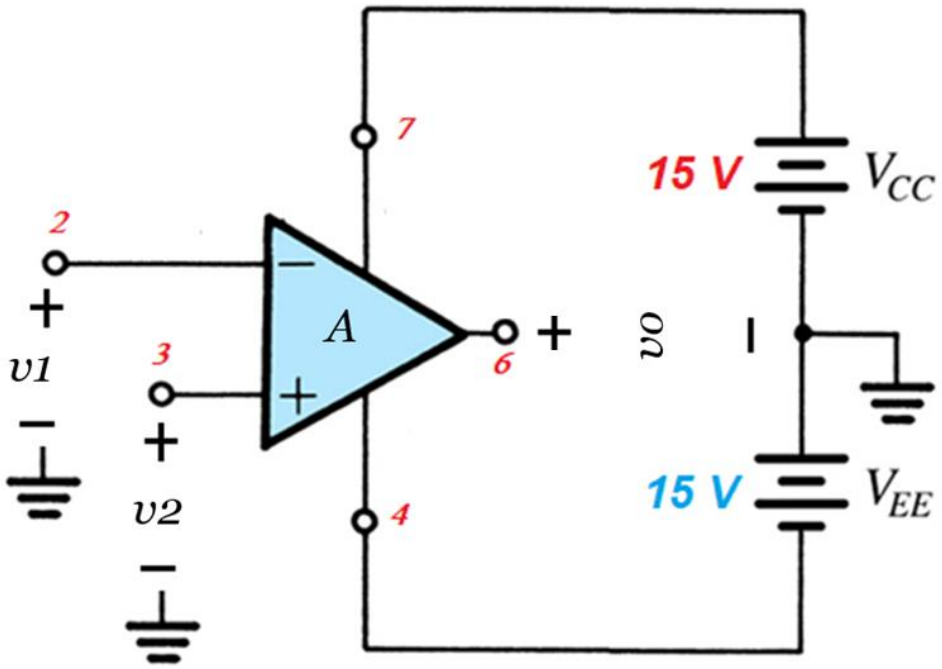
$$A_{vo} = \frac{v_o}{v_i} > G = \frac{v_{out}}{v_s}$$

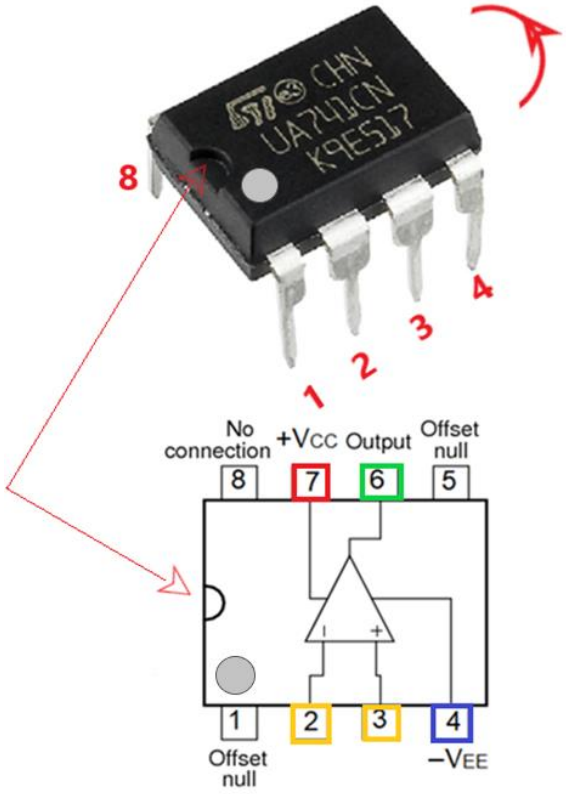
amplifier voltage gain — *overall voltage gain*

What is an *Op*erational *Amp*lifier (Op Amp)?

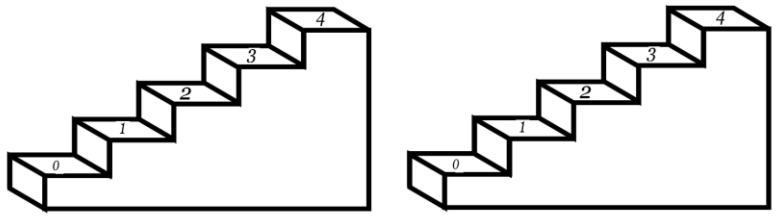
- Versatile *integrated circuit* (IC).
- Performs various analog signal processing tasks.
- Amplifies the difference between two input voltages ⇒ eliminates input signals common noise.

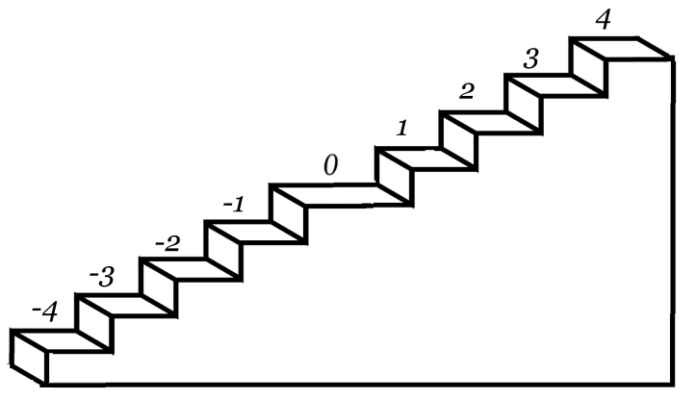
Symbol and Power Connections



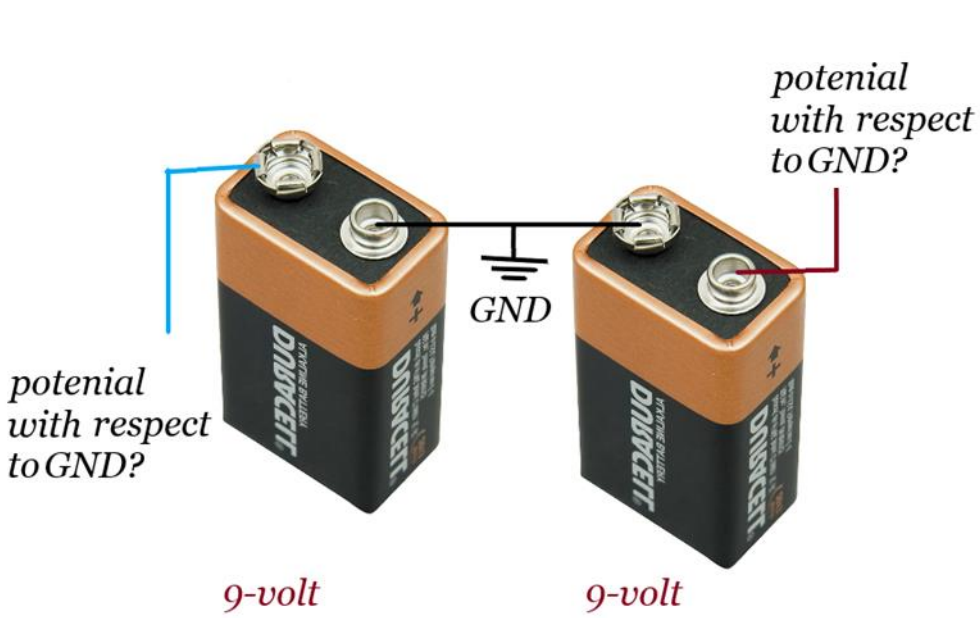


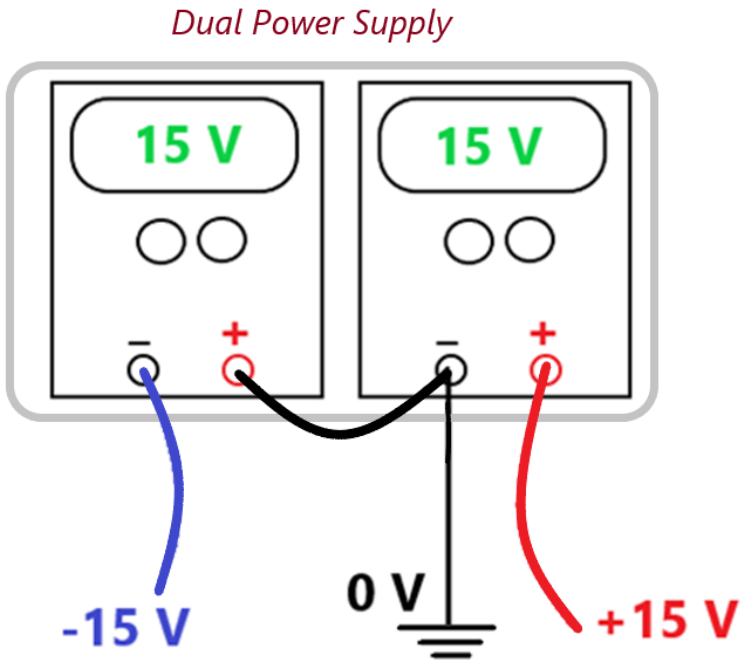
Stairs Analogy with Potential





Creating Bipolar Supply !

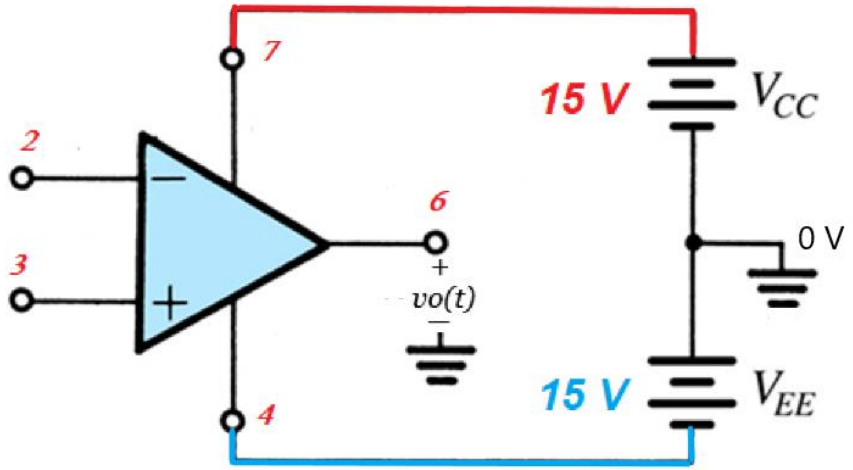


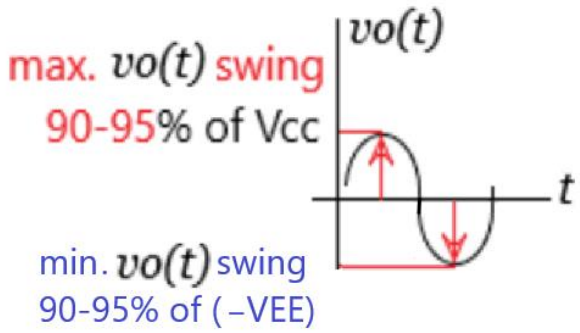


Power Rails Limit The Output Swing Range

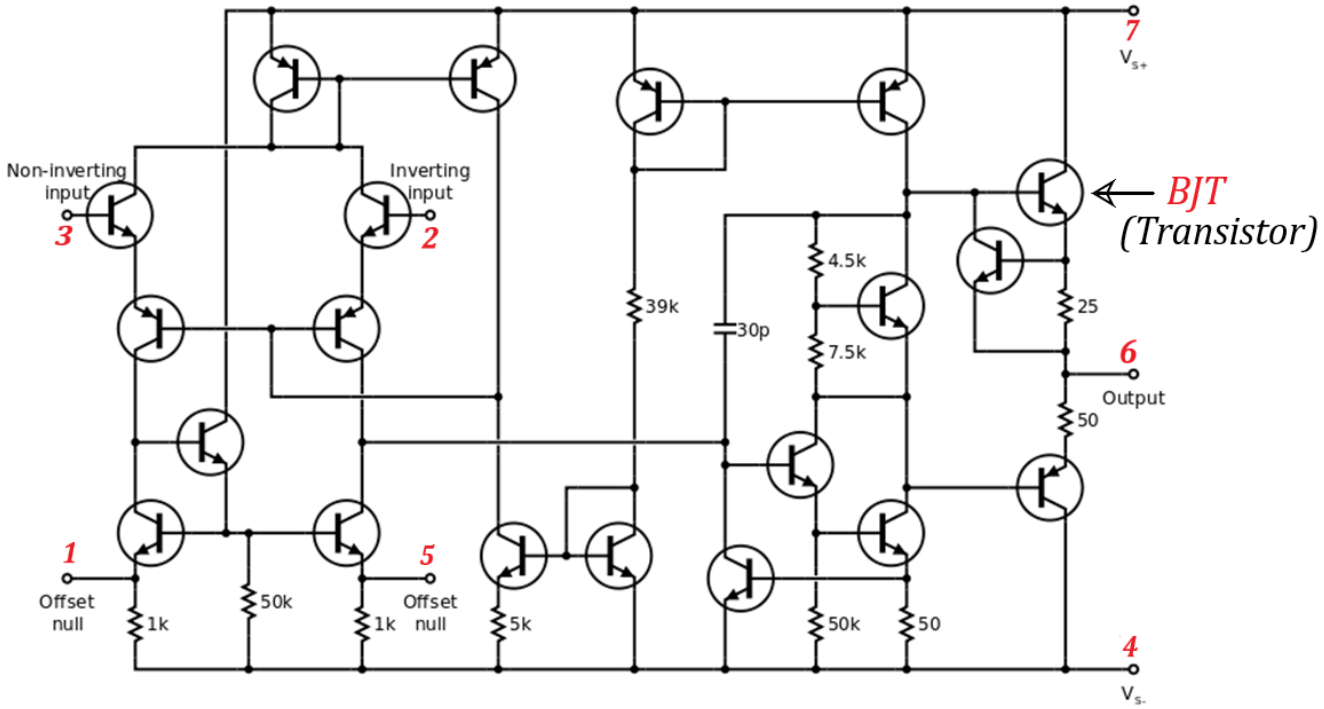
The power supplies bias the internal transistors of the op amp, and provide power to the output load and the feedback network.

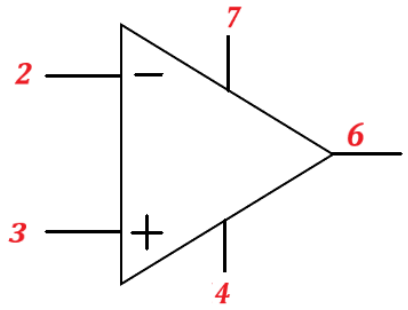
The output voltage is limited at about 90 to 95% of the power supplies voltages.



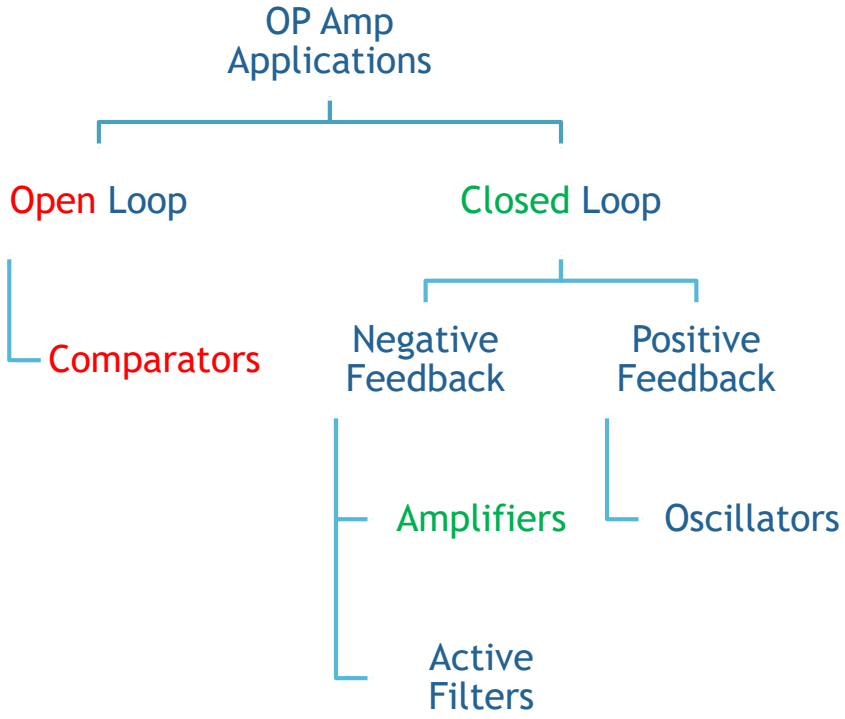


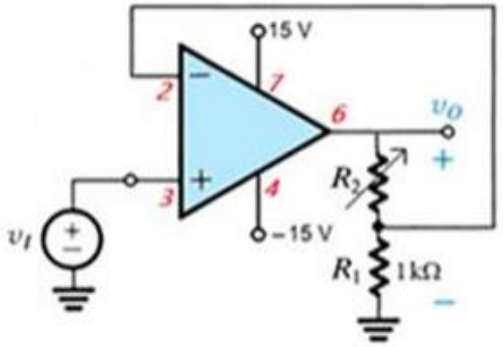
Internal Circuitry of the 741 Op Amp



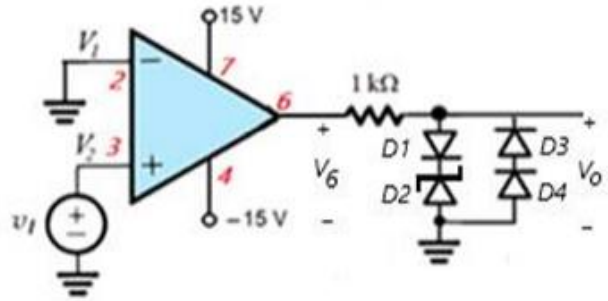


Op Amp Main Applications





Non-inverting Amplifier

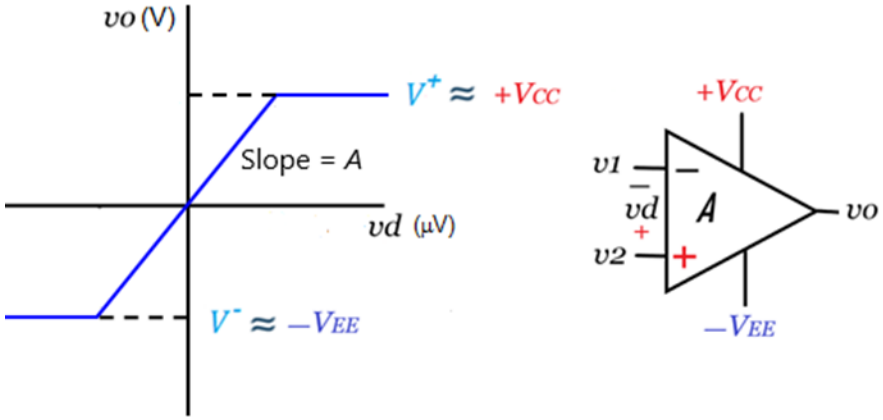


Comparator

Linear Operation

As long as an op amp output is not saturated at its power rails limits, it operates linearly:

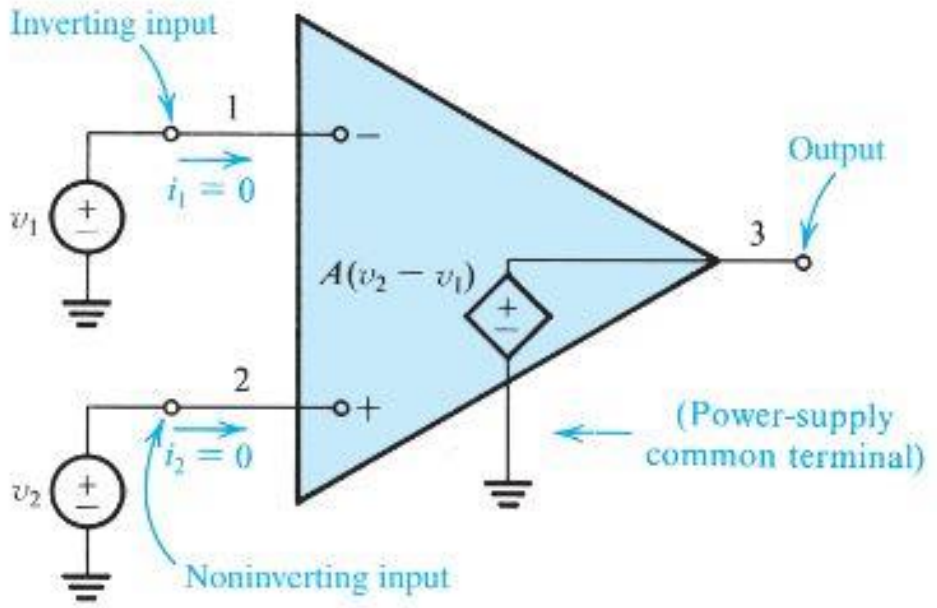
$$v_o(t) = A \times [v_2(t) - v_1(t)]$$



Due to very large A, the linear region has the virtual short circuit property,

$$v_2(t) \approx v_1(t)$$

Equivalent Circuit of the Ideal Op Amp In the Linear Region



igure 2.3 Equivalent circuit of the ideal op amp.

Why Do We Need Negative Feedback ?

- To obtain a controllable and predictable **closed-loop gain *G***
- Negative feedback makes the closed-loop gain **G** almost entirely dependent on external components (e.g., R_1 and R_2) rather than on ***A***, the Op Amp gain.

The Non-Inverting Amplifier

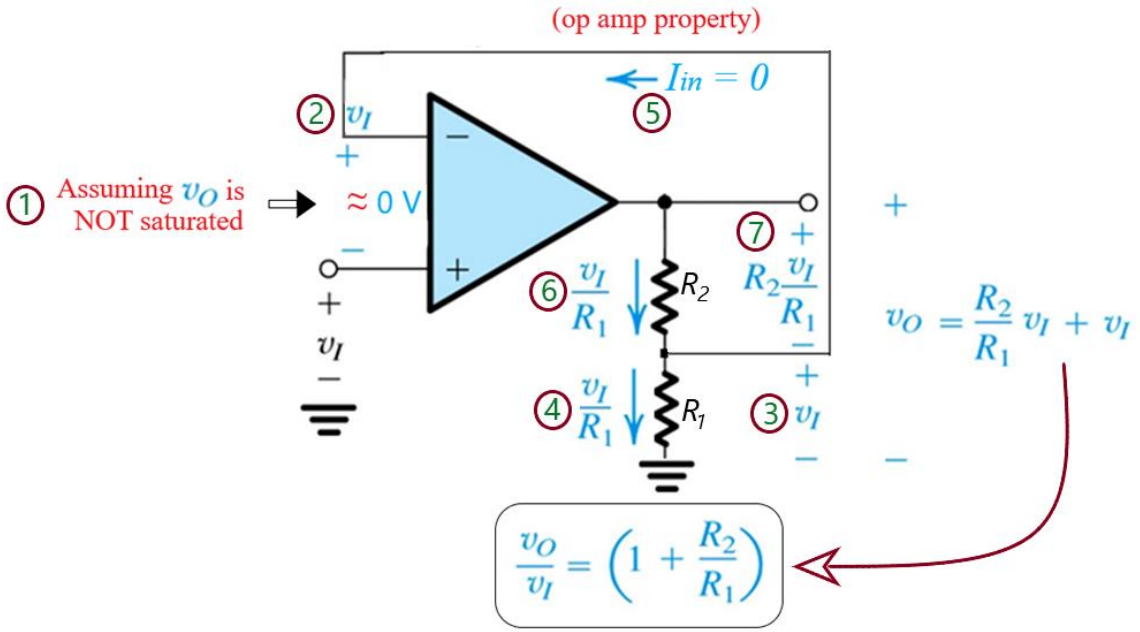
Presuming linear region operation:

$v_2 \approx v_1 \approx v_I$

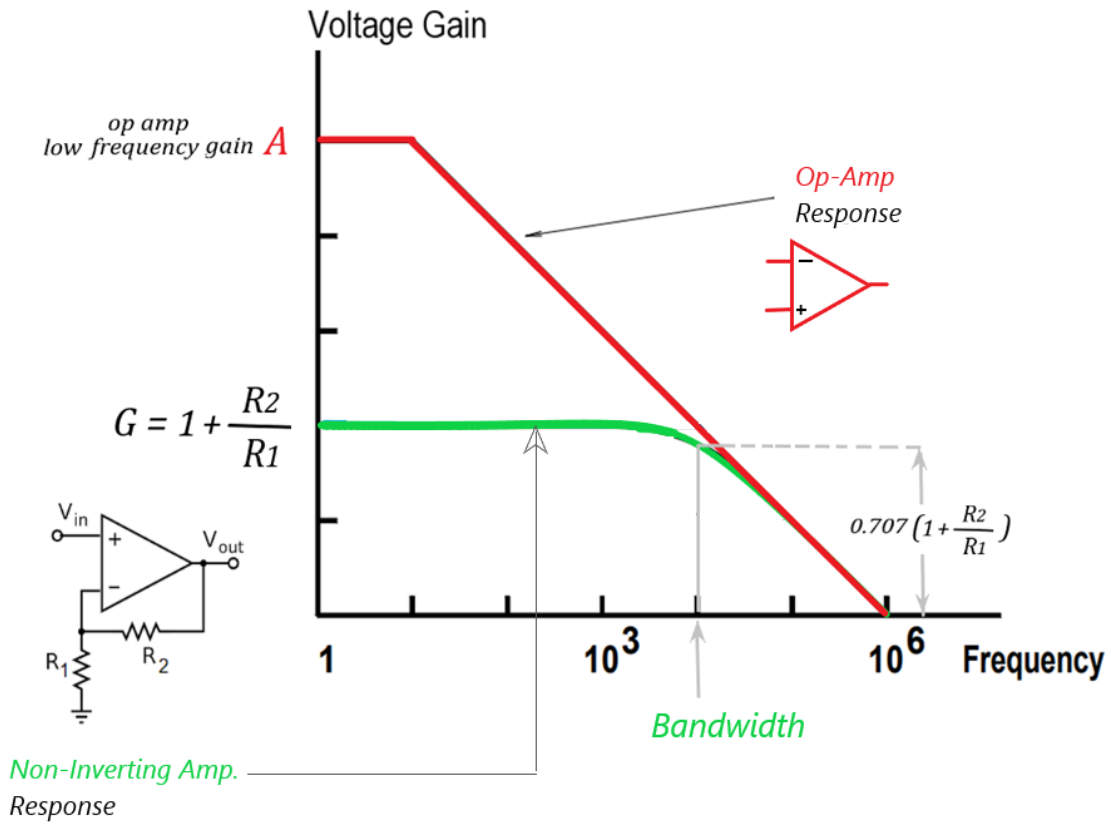
Since the noninverting terminal of the op amp draws negligible current, R_1 and R_2 are effectively in series:

$i_{R2} = i_{R1} = v_I / R1$

$$G \equiv \frac{v_o}{v_i} = 1 + \frac{R2}{R1}$$



Bandwidth of an Amplifier



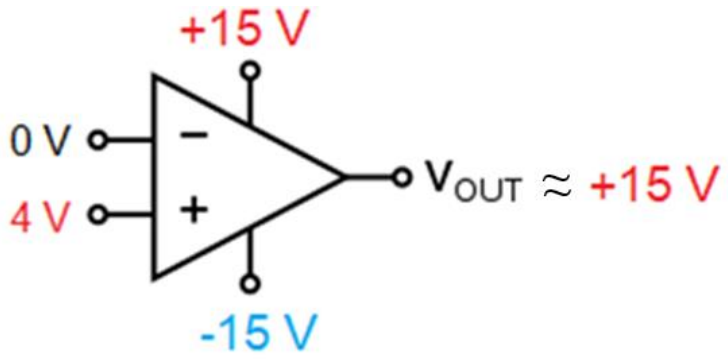
The Op Amp as a Comparator

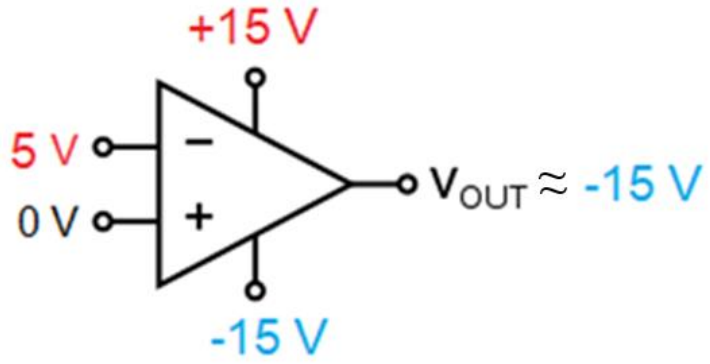
If no negative feedback is employed, an op amp is said to operate in the **open-loop** mode.

In that mode, an op amp basically works as a comparator.

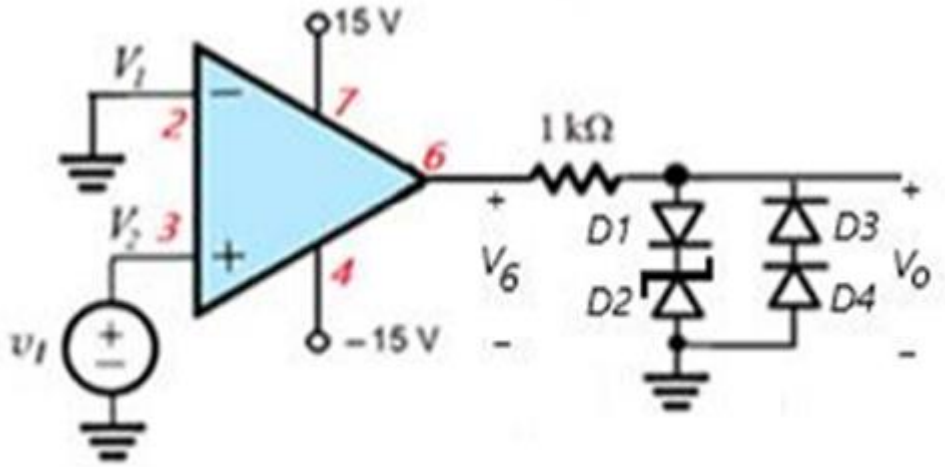
If v_2 exceeds v_1 by few tens of microvolts, the output will saturate at an upper saturation level $V_O^+ \approx V_{CC}$.

If, however, v_1 exceeds v_2 by few tens of microvolts, the output will saturate at a lower saturation level $V_O^- \approx -V_{EE}$.





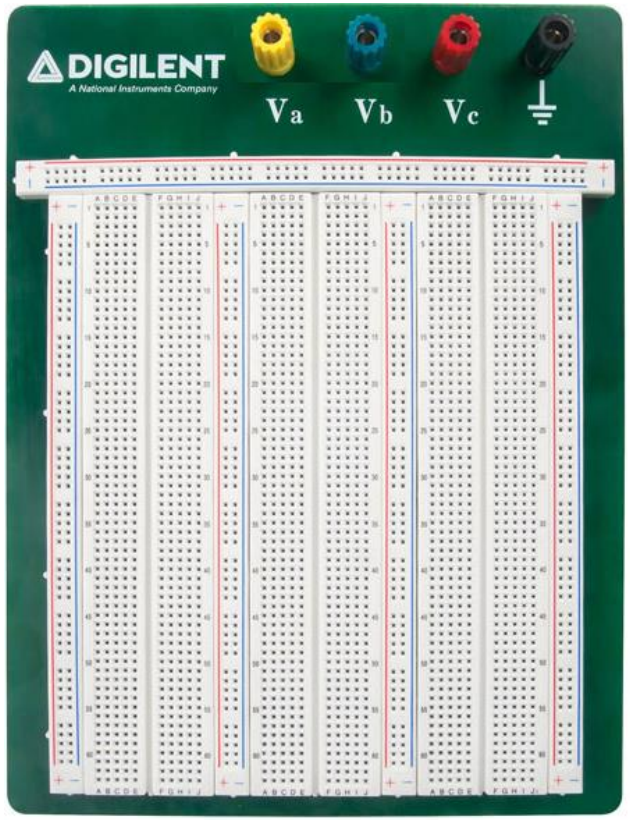
Creating a Square Wave Using Op-Amp Operating as a Comparator!



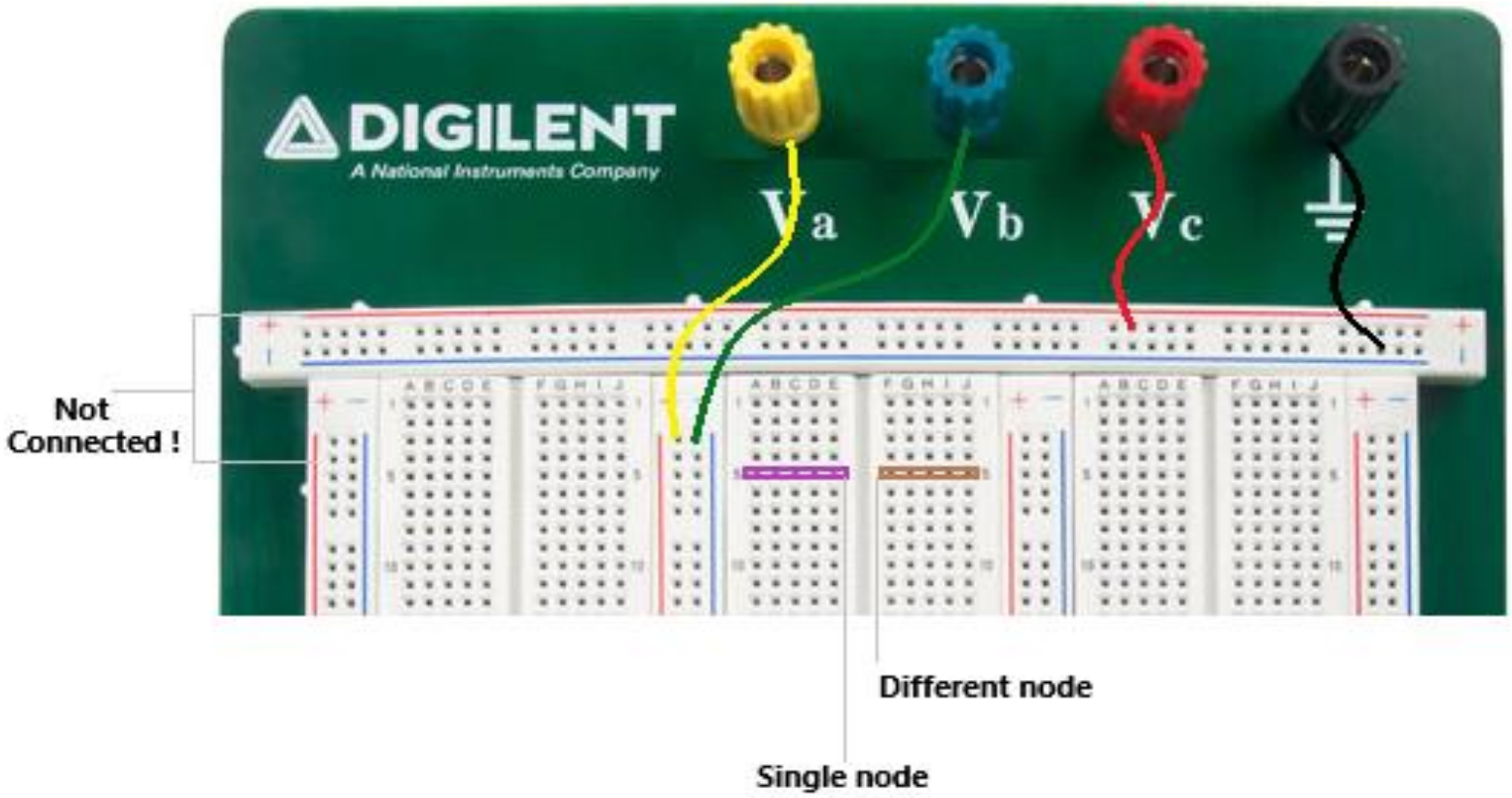
Comparator

Procedure A

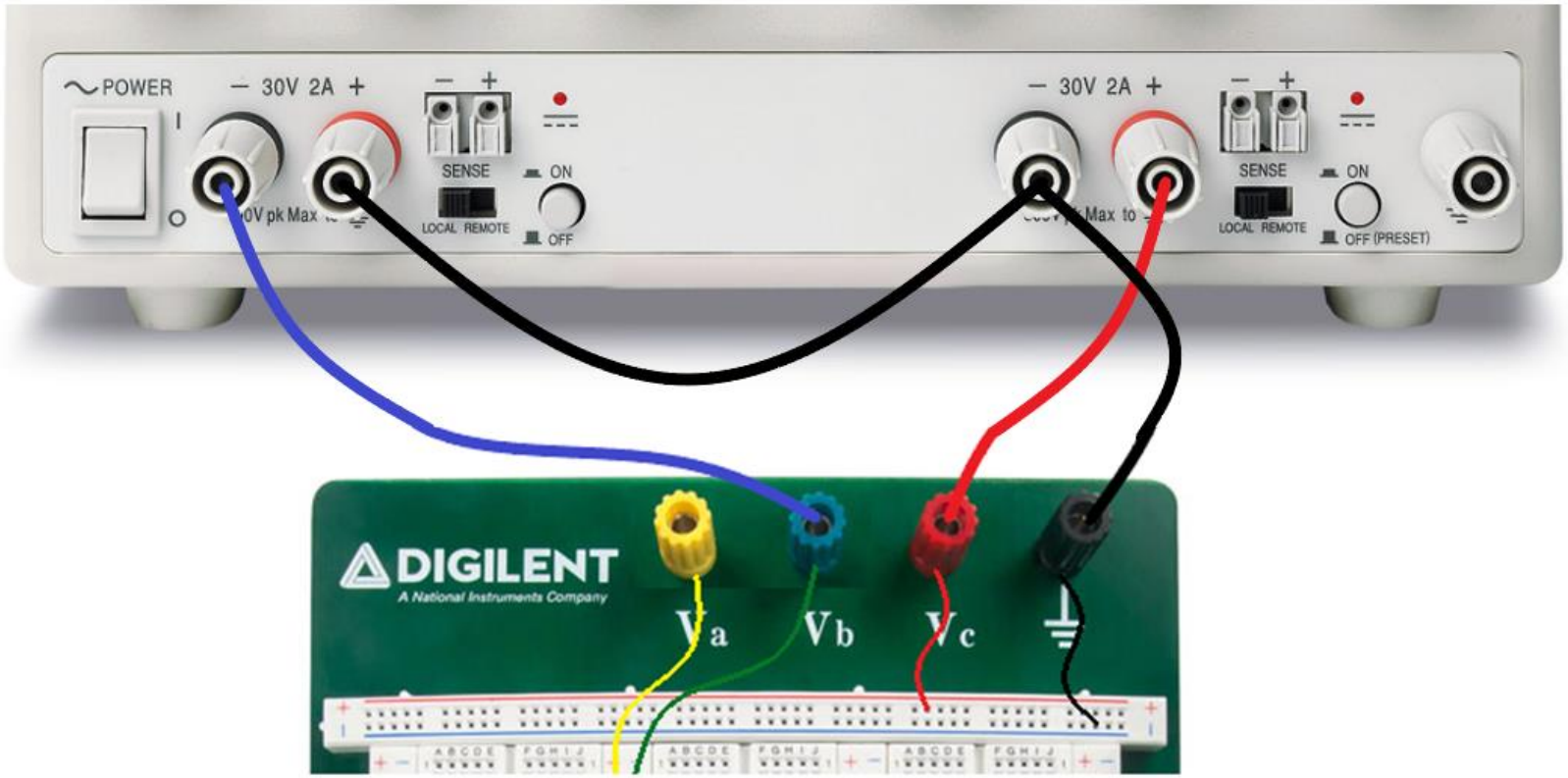
Breadboard Basics 1/4



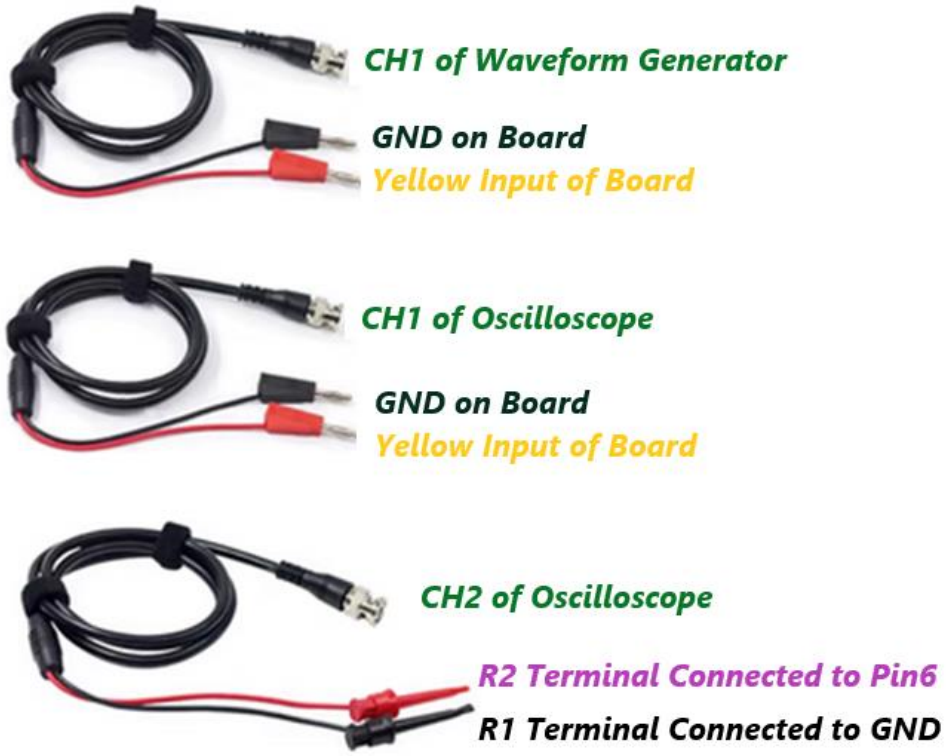
Breadboard Basics 2/2



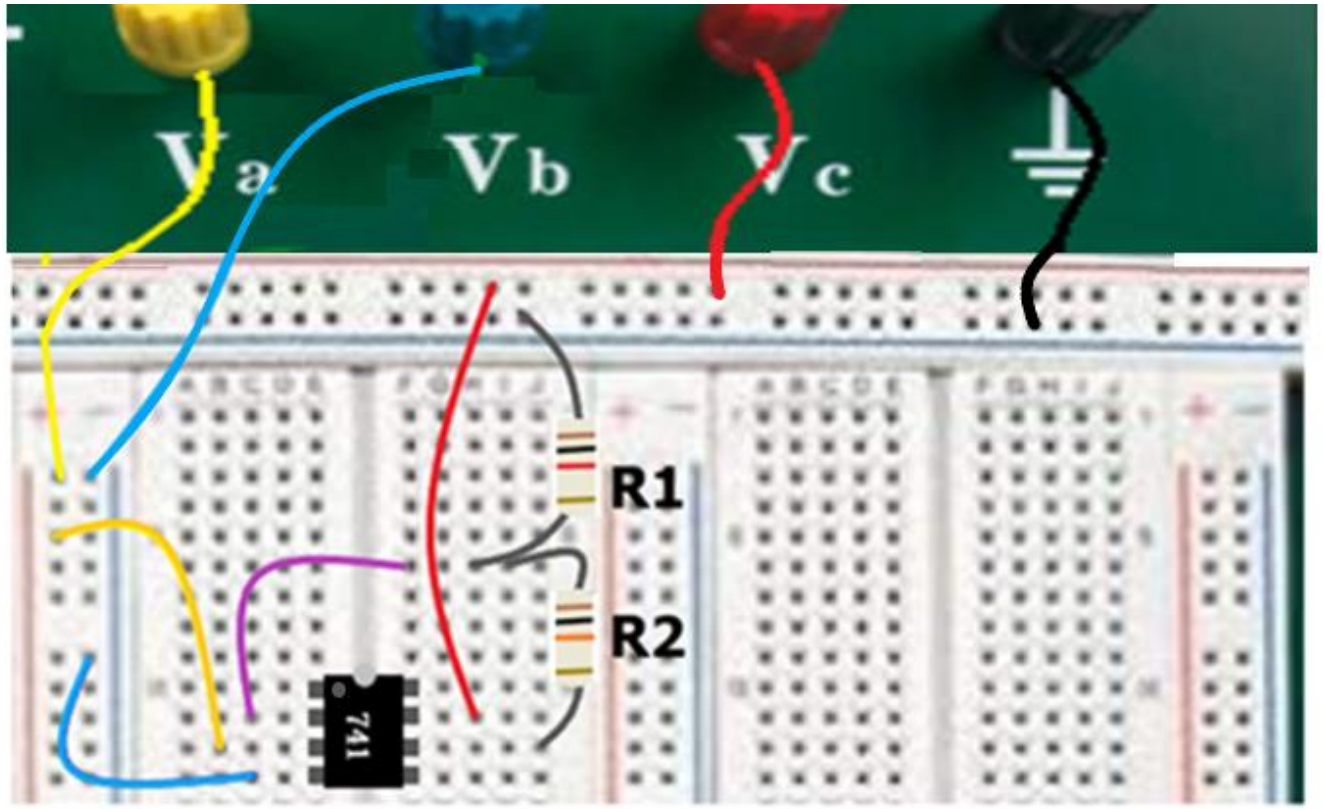
Power Supply Connections



Coaxial Cable Probes Connections



Non-Inverting Amp (Breadboard Version)



The End

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